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Special issue article

The role of advisory provision and policy objective information in dairy farmers' preferences for methane-reducing feed additives

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Abstract

In a discrete choice experiment with 320 Swedish dairy farmers, we investigate preferences for methane-reducing feed additives, finding that higher environmental effectiveness and a farm advisor as the provider of accompanying information increase the probability of adoption. Additionally, an information treatment on methane reduction policy objectives lowers willingness to accept for the 30 per cent methane-reduction level of the feed additive. Latent class analysis shows that innovation-friendly farmers are more likely to be adopters, while those with stronger pro-animal-welfare attitudes are not. Finally, we predict feed additive market shares and discuss the economic implications for emissions reduction.

Keywords: feed additive; 3-nitrooxypropanol (3-NOP); climate change; discrete choice experiment; information provision; greenhouse gases.

JEL classification: Q18, Q54, C35

1. Introduction

Reducing methane emissions from agriculture is necessary for combating climate change (Matthews, 2023). Methane has a much higher global warming potential than carbon dioxide (CO_2) but breaks down more quickly in the atmosphere, making it relevant for short-term climate action (Dijkstra *et al.*, 2025). Several new and established technologies aim to reduce methane from enteric fermentation (Arndt *et al.*, 2022), including feed efficiency improvements, breeding, pasture feeding and manure management. Innova-

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tions such as antimethanogenic feed additives—3-nitrooxypropanol (3-NOP), nitrate and red algae—are being widely tested for their methane-reducing effects in trials (Kjeldsen *et al.*, 2023; Dijkstra *et al.*, 2025). Uncertainties remain regarding the long-term impacts, potential side effects and trade-offs of feed additives (Kjeldsen *et al.*, 2023; Gastelen *et al.*, 2024), including how to estimate and model them (Dijkstra *et al.*, 2025). Denmark has mandated the use of 3-NOP feed additives or high-fat feeds in dairy farms for a minimum of 80 days in 2025, compensating the cost of the additive, with additional subsidies being available for exceeding the minimum requirements (Danish Agricultural Agency, n.d.). In neighbouring Sweden, dairy processors seeking to meet science-based targets for greenhouse gas (GHG) emission reduction (Gottlieb and Edenbrandt, 2024) have also been exploring methane-reducing feed additives and remuneration to farmers for GHG mitigation. In the USA, beef cattle producers are not yet allowed to use 3-NOP additives, but Luke and Tonsor (2024) find that processor premiums would be preferred over subsidies. The emerging nature and claimed potential of these innovations highlight the need to understand farmers' preferences and willingness-to-accept (WTA) payments for implementing methane-reducing feed additives and the overall uptake and methane-reduction potential. Given remaining uncertainties farmers may have regarding the perceived effectiveness of the additives (Sidhoum *et al.*, 2025) and the rapidly emerging policies in this area, we conducted a discrete choice experiment (DCE) with 320 Swedish dairy farmers to investigate their preference for adopting 3-NOP feed additives with varying methane-reduction potential, provider and frequency of accompanying advisory information, as well as bonus payments from the processors.

We conceptualize farmer innovation adoption decisions to be shaped, among other things, by the availability of information that can update farmers' prior beliefs and accompanying advisory information that can offer learning and improvements throughout implementation. Information therefore spans various layers—from the farmer, their closer professional network, to the broader societal and policy context. First, relating to information received in advisory context, prior literature indicates that some advisory support, like complementary digital advisory information (Kirchner and Musshoff, 2024) or free start-up consultancy (Zemo and Termansen, 2018), is desirable by farmers when adopting new or environmental practices. At the same time, ex-post analysis indicates that advisory participation in farmer discussion groups does not lead to the actual uptake of environmental practices (Balaine *et al.*, 2023). The timeliness of advisory services is also relevant to farmers, as demonstrated in their willingness to take up precision farming technologies (Späti *et al.*, 2022). To address the first aspect of information provision, in terms of advisory support for innovation adoption, we incorporate DCE attributes on who provides the advice on feed additives and how often.

Second, in terms of information provision treatment effects in DCEs, the evidence is still inconclusive, demonstrating positive, negative and non-significant impacts (e.g. Van Loo *et al.*, 2020; Czajkowski *et al.*, 2021; Welling *et al.*, 2023; Oyinbo and Hansson, 2024). In these studies, the focus is of-

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ten on the content of the provided information, such as the benefits of a practice (Oyinbo and Hansson, 2024), the identity of those consuming the product (Schwartz *et al.*, 2020), or how information is communicated (Yang and Hobbs, 2020). While the impacts or benefits are the most prevalent aspect of the content of information provision, we focus on testing the role of providing information about the policy context and the relevant objectives or goals that have proliferated recently. Some examples include the Global Methane Pledge (GMP), the European Union's (EU) carbon neutrality goals and rapid uptake of corporate science-based targets for GHG emission reduction. We thus introduce an information provision treatment about the existence of public (e.g. national) and private (e.g. corporate) GHG and methane reduction objectives to test whether such policy objective information affects the WTA emission reduction potential of the additive and the choice of opting out of this innovation. In doing so, we add to the literature on farmer preferences and choices about two types of information: advisory information provision conditional on acceptance of the new practice and policy information treatment before choices. The findings highlight the relevance of the advice providers and point to modest effects on preferences for methane-reduction effectiveness from a single short exposure to information on policy targets for methane reduction.

Finally, since a broad set of farmers' behavioural factors is important for preferences (Dessart *et al.*, 2019; Wuepper *et al.*, 2023), choices and susceptibility to new information around environmental and climate change mitigation measures, including feed additives (Tarruella *et al.*, 2025), we follow the increasing practice of including them in stated preference studies (Muzzillo *et al.*, 2024). From innovativeness and farm animal welfare (FAW) attitudes to factors affecting the feasibility of adoption, we explore their role in the heterogeneity of farmers' potential uptake of a methane-oriented innovation and their preferences for advice.

The paper proceeds by positioning the study in the literature and conceptual framework in Section 2. Section 3 explains the experimental design and treatment. Section 4 offers and discusses results. We conclude in Section 5.

2. Conceptual framework, hypotheses and previous literature

We develop a conceptual model that builds on the idea that the decision to adopt feed additives is multi-layered (see Fig. 1). Layers interact and influence the decision to adopt or not to adopt feed additives on the farm. In Sections 2.1–2.3, we provide detailed motivation for each of the three layers in the framework. We begin with the advisory and support layer, followed by the policy information layers, which are more central to the focus of this paper and for which we develop hypotheses. Afterwards, we briefly introduce the innermost layer of the farmer and the processor, which is related to our heterogeneity analysis and the payment attribute.

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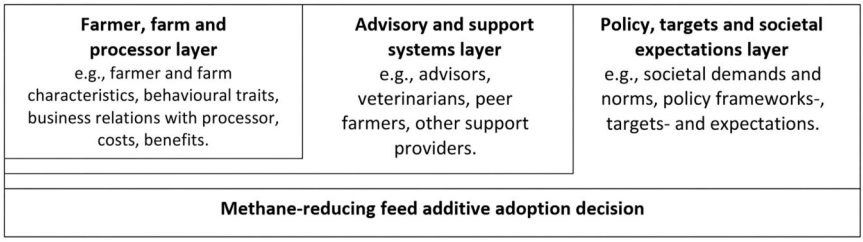

Fig. 1. Conceptual framework.

2.1 Advisory and support systems

Farmers' objective knowledge positively impacts the stated uptake of agri-environmental schemes (Czajkowski *et al.*, 2021). Building on the theoretical framework of Oyinbo and Hansson (2024), we base the analysis on an expected utility model of farmers' preferences for adopting feed additives that lower GHG emissions (details in Section A of the Supplementary Appendix). The model assumes that farmers rely on their prior beliefs about these practices when faced with uncertainty or misconceptions about the potential effectiveness, benefits, costs and risks of adopting feed additives—often due to limited access to relevant decision-making information. As new information is relevant, the model accounts for it in two ways, as an attribute of the product, service or practice on offer, in line with advisory aspects considered by DCE studies (e.g. Zemo and Termansen, 2018) and as ex-ante information provided to a sub-sample of respondents in line with the information provision treatment literature (e.g. Oyinbo and Hansson, 2024). The first component is information obtained conditional on initiated adoption, that is, advisory information provision. It is a component of the innovation adoption proposition in the form of accompanying and potentially recurring advice. The presence of advisory, technical and administrative support has been identified as an important aspect of farmers' willingness to adopt agri-environmental practices and contracts (Czajkowski *et al.*, 2021; Schulze *et al.*, 2024). Several stated preference studies have considered the presence or complementarity of agricultural knowledge and support systems. Some studies have investigated the impact of the general presence of free advice (Schreiner and Latacz-Lohmann, 2015), free start-up consultancy (Zemo and Termansen, 2018), or complementary information services such as weather forecasts (Kirchner and Musshoff, 2024) on the adoption of farming practices. Other studies have been taking into account the participatory decision-making in advisory services (Tesfaye *et al.*, 2019), the adoption of support systems to sites (Oyinbo *et al.*, 2019) or farming types (Otieno *et al.*, 2011), as well as information channels through which advice is offered (Altobelli *et al.*, 2018).

In the context of methane-reducing feed additives, Tynan *et al.* (2023) explore farmers' views in the USA towards algae-based alternatives. They emphasize the importance of advice provider and farmers' trust in the advisor.

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Specifically, they found all farmers to indicate most trust towards their nutritionists, followed by seeking advice from extension specialists, veterinarians and other farmers for information about feed supplements (ibid.). It is thus relevant to consider the preferences towards the advice provider as potentially relevant for increasing innovation uptake. Given the expected positive preferences for some advice vs. no advice in the above-mentioned overall literature on advice provision, we posit that:

H1 : Farmers will have a positive preference for all types of advice providers on feed additives compared to no advice.

The timing of receiving technical support following technical difficulties was also relevant for farmers' adoption of precision farming technologies (Späti *et al.*, 2022). However, empirical findings from the rolled-out extension services suggest that among dairy farmers exposed to advisory for a longer period, the effect of farmer discussion groups with advisory supervision on the implementation of sustainable management practices, economic and sustainability outcomes was insignificant, mixed and heterogeneous (Balaine *et al.*, 2023). The duration of exposure to advisory services is thus a potentially relevant issue to consider even in our case, given the uncertainties associated with the longer-term effects of the feed additives on the cows and emission-reduction efficiency (Kjeldsen *et al.*, 2023). Assuming that farmers would derive utility from more and updated information throughout the rollout of feed additives, we test:

H2 : Farmers will have a positive preference for more frequent advice.

2.2 Policy, targets and societal expectations

The second type of ex-ante information treatment has become an area of substantial interest, in part due to the relatively low cost and ease of information provision. The provided information often captures the issues of broader social and environmental relevance, such as the environmental benefits of agri-environmental schemes (Czajkowski *et al.*, 2021), 'balanced' (Oyinbo and Hansson, 2024) or 'personalized' (Zachmann *et al.*, 2023) information combining economic, environmental and social implications of sustainable farming practices or environmental, FAW, brand and technical aspects of an innovative beef product (Van Loo *et al.*, 2020).

In many areas of improving farm sustainability outcomes, it takes some time from the initial planning of public and private policy objectives until practical adoption possibilities and monetary incentives become available to farmers. Specific policy ambitions and commitments then diffuse through advisors to farmers, or farmers directly learn about new policies and societal expectations in the outer layer of our framework.

Regarding the recent emphasis on methane, over 150 countries, including Sweden and the EU, have signed up to the GMP goal of reducing anthropogenic methane emissions by at least 30 per cent by 2030 compared

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to 2020. With the agricultural sector accounting for a staggering 56 per cent (12 per cent) of methane (all GHGs) emissions in the EU as of 2022 (European Environmental Agency, 2025), the EU now has a methane strategy, which specifies several cross-sectoral and sector-specific measures to improve the measurement, reporting and verification of methane emissions, as well as to reduce these emissions. Eco-schemes of the Common Agricultural Policy are expected to reward farmers for mitigation practices, including methane-reducing feed additives (Matthews, 2023). As of January 2025, Denmark began to mandate the use of 3-NOP or high-fat feed in conventional farms with over 50 dairy cows for 80 days, with the additive's cost compensated and an additional subsidy available for all-year use (Danish Agricultural Agency, n.d.). Swedish policy-makers have also reviewed feed additives as a possible measure to achieve the GMP and Swedish climate goals (Swedish Environmental Protection Agency, 2023). Corporate demand for and supply of methane-reducing feed additives for agriculture is also increasing. Nordic dairy processors in Sweden, Denmark and Finland have science-based targets to reduce their Scope 3 (i.e. supply chain and on-farm) GHG emissions. Some of them have also been widely testing the feed additive, and one processor has piloted a new brand of milk that claimed 25 per cent lower GHG emissions, partly due to the use of the feed additive. Despite these pilots, most Swedish farmers have not been offered the option to use these yet (in our sample, only 3 per cent have tested feed additives).

Related to the outer layer of our framework, we explore whether a single-time provision of information to a randomized half of the sample about these climate policy objectives and goals is relevant for farmers' preferences for methane-effectiveness of the additive and opt-in overall. The model discussed in Section 2.2. and Section A of the Supplementary Appendix, therefore, incorporates Bayes' theorem to examine how providing additional information about methane objectives can influence farmers' marginal utility of the additive's methane reduction effectiveness. Overall, we expect the information treatment to increase the perceived utility of the feed additive methane-reduction attribute and decrease the utility of the opt-out. Information can 'increase the awareness of the relevance of one's behaviour and thus their intended environmental action (the sense of control), and through these channels influence pro-environmental behaviour' (Czajkowski *et al.*, 2021). If information influences the awareness of the relevance of additives as a pro-environmental measure, likely increasing the probability of a good state in our conceptual model, with more monetary and non-monetary benefits, farmers are likely to view the additives and their mitigation potential more positively. This would, in turn, imply that they require lower compensation for adopting such an innovation, specifically additives that offer higher and more socially desirable emission reduction. Accordingly, we formulate the following hypotheses:

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H3 : Farmers exposed to the policy objective information will have a more positive preference and a lower WTA for the methane-reduction attribute of feed additives.

H4 : Farmers exposed to the policy objective information will have a less positive preference and a higher WTA for keeping the status quo.

2.3 Farm, farmer and processor

This part of the framework deals with the material payoffs, farm and farmer characteristics and behavioural factors that influence farmers' preferences and potential adoption of feed additives. Farmers consider both pecuniary and non-pecuniary benefits—such as bonus payments from the dairy company and expected methane reduction levels—as part of their decision-making. Previous studies highlight that behavioural factors, including cognitive and dispositional ones, influence farmers' agri-environmental practice adoption (Dessart *et al.*, 2019; Barreiro-Hurle *et al.*, 2023; Schaub *et al.*, 2023; Barreiro-Hurle and Rommel, 2025). Uncertainty and prior knowledge are particularly relevant for preferences when novel sustainable technologies are adopted (e.g. Faccioli *et al.*, 2019; Ortega *et al.*, 2020; Oyinbo and Hansson, 2024). Regarding views on antimethanogenic feed additives among Finnish dairy farmers, uncertainty has been most pronounced regarding their methane mitigation potential (Sidhoum *et al.*, 2025), potentially since methane reduction is assessed through a formula rather than measured directly. Besides the methane efficiency attribute, we thus include variables on farmers' stated familiarity with feed additives and the brand Bovaer in particular (all behaviourally related measures are outlined in the Section D of the Supplementary Appendix). We also assess farmers' actual knowledge of emissions from enteric fermentation.

Farmers assess the uncertain benefits and risks based on the information they get access to, including through learning by doing or from peers or advisors (Chavas and Nauges, 2020). The interaction of cognitive ability and receptiveness to advice shapes adoption dynamics: while cognitive ability speeds adoption as expected, being receptive to advice slows adoption for farmers with high cognitive ability (good at learning by doing), since they benefit from waiting to incorporate additional advice (Barham *et al.*, 2018). Dispositional factors, such as perceived risks in livestock management, also concern FAW. For example, Tynan *et al.* (2023) show that farmers voice frustration with attributing blame for GHG emissions to the farm or specific farm types like pasture-based systems. They argue that farmers' decisions about feeding algae-based feed supplements depend on perceived benefits for herd health, cow behaviour, availability of alternatives in organic farming and cost. Trust towards one's contacts and sources of information is likely to matter for willingness to act upon advice and take up innovations (Arbuckle *et al.*, 2015; Strazzera *et al.*, 2022). Based on these considerations, we include measures of trust in general and towards specific stakeholders (Haerpfer *et al.*, 2022), risk aversion (Dohmen *et al.*, 2011), as well as openness to innovation

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(Bougherara *et al.*, 2023). Our measure of trust towards specific stakeholders is operationalized as confidence in different sources of information, which may be correlated with receptiveness to advice. We measure FAW through scales that benchmark the well-being of animals against profitability, working conditions and the environment.

Finally, various attitudes are relevant for climate change mitigation practices in agriculture, including climate change and environmental attitudes (Kreft *et al.*, 2021) and a precautionary attitude (King, 2023). Related to these and motivated by the attribution theory (Jang, 2013), we consider climate change beliefs of farmers (Islam *et al.*, 2013), specifically the extent to which farmers attribute GHG emission reduction interventions to concern the dairy sector and cows. In line with Czajkowski *et al.* (2021), we account for potential path dependency through variables indicating whether farmers have adopted other sustainable feeding practices encouraged by processors. To capture adoption feasibility, we include whether the farm is organic or in conversion (and thus currently restricted from using 3-NOP), whether the farmer owns a feed mixer (required for application), whether manure is used for biogas production (a possible constraint), and which dairy processor the farmer supplies (relevant as not all processors have piloted the additive yet).

These farm and farmer characteristics, guided by the framework, allow us to explore preference heterogeneity for feed additives across behavioural and structural dimensions.

3. Methodology and data

3.1 Choice scenario, attributes and validity

The choice experiment design was discussed in three workshops with practitioners and processors and in interviews with farmers and researchers of feed additives. We qualitatively pretested the survey with a 'read aloud' option to ensure comprehension, credibility, accuracy and appropriateness of the survey. A pilot with 57 dairy farmers was conducted to obtain prior estimates for the choice design (Johnston *et al.*, 2017). The data, code and pre-registration plan are available on OSF.[1] The attributes and levels are displayed in Table 1.

The choice scenario explained to farmers that their dairy company would offer them feed additives for their dairy herd free of charge. In addition, they would receive a bonus payment per kg of milk. Feed additives could also come with free-of-charge advice, and the effectiveness of feed additives could differ. Section B of the Supplementary Appendix shows how the choice scenario and attributes were presented.

1 The pre-registration project files can be found here https://osf.io/uq582/. Two additional hypotheses relating to the advisory attributes were requested to be made specific in the article during review process and are thus not available in the plan, besides the used priors for their estimates and thus expected direction of preferences. Our hypothesis 4 has also been corrected in the review with regard to the direction of the effect for WTA, despite the correct underlying assumption of a reduced preference for status quo effects of treatment.

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Table 1. Attributes and levels of the DCE.

Attribute name	Levels
Who provides advice	No advice offered (*reference*), farm advisor, Veterinarian, experienced farmer
How often advice is provided	0 (No advice offered), 1 (Once per year), 2 (Twice per year), 3 (Three times per year)
Methane emission reduction	10 per cent less (*reference*), 20 per cent less, 30 per cent less, 40 per cent less
Milk price bonus, öre per kg of milk	0–35 öre in intervals of 5 öre (ca. 0–3.5 Eurocents per kg of milk in increments of 0.5 Eurocents)

The first attribute on *advice provision* reflects the option provided during pilots of feed additives in Sweden and potential alternatives to the current rollout of dairy companies. In pilots, the farm advisor offered information regarding the use of the additive, discussed the dose, ensured fit to the feed ration and was involved in follow-ups with farmers. Given the articulated and potential concerns regarding cow health and welfare effects, we also consider a veterinarian as the provider of advice, as well as peer-to-peer advice from farmers experienced in the application of the feed additive, reflecting peer-based extension services (e.g. Balaine *et al.*, 2023).

The second advisory attribute—*frequency of advice*—aims to capture the preference for frequency as a proxy for the amount and novelty of information provided. We presented the attribute as such that 'can include advice on how to start using the feed additive or a follow-up discussion of any questions' of the farmer, and where on each occasion, the farmer 'would be able to discuss the latest available information'.

We also study the *methane emission reduction* attribute of the feed additives with levels that are informed by the upper bound of estimated reduction for 3-NOP feed additive across studies with different doses, feed rations and cow characteristics (Kjeldsen *et al.*, 2023), including the current commercial claim for Bovaer of an approximate 30 per cent reduction. We chose not to include uncertainty of methane reduction properties or other potential side effects of the feed additives in the choice design to reduce complexity and the number of attributes, in favour of average expected methane-reduction effectiveness as an area of special uncertainty to farmers (Sidhoum *et al.*, 2025). This attribute relates to a farmer's pro-environmental attitudes in the innermost layer of our framework while interacting with policy level.

The *milk price bonus* attribute reflects additional payments beyond the cost of the additive by the dairy company. Levels are adjusted to the Swedish context and designed to offer a significant variation. This attribute strongly affects costs and benefits in the innermost layer of our framework. An attribute level of zero implies that the farmer would apply feed additives with only its cost covered and no additional payment. However, interviews with farmers and dairy industry representatives suggest that farmers may still require a financial

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incentive[2] to use the feed additive, and previous findings suggest that farmers want more than direct costs to be covered (Christensen *et al.*, 2011). The cost for the approved in the EU feed additive has been estimated at 1 Eurocent per kg of milk (approximately 12 Swedish öre at the time of data collection).

We followed best practice regarding the validity of DCEs (Johnston *et al.*, 2017), including a cheap talk script for each choice card. An example of an English translation of a choice card with the script is presented in Section B of the Supplementary Appendix. Policy context was made credible and consequential (Welling *et al.*, 2022) by explaining that results will be communicated to dairy companies to be taken into consideration for incentive design. When assessing how respondents perceived consequentiality (Zawojska *et al.*, 2024), we find that 32 per cent of respondents stated they believed that the findings will be considered by the dairies "very much or much", 31 per cent thought 'somewhat' and 21 per cent thought 'not at all'.

3.2 Statistical DCE design

We assumed random utility maximization with linearly additive attributes. The design obtained in Ngene was a D-efficient design with 24 choice situations split into four blocks, modelled for a multinomial logit (MNL) model. We included a constraint that when advice frequency level is none, no one would be providing it and vice versa, allowing all the other combinations to be unconstrained. Note that we rely heavily on mixed logit (MXL) models in our results. However, as the priors come from a small sample, and the MXL model has more parameters, we used the simpler MNL model to guide our experimental design simulations.

The choice cards included unlabelled alternatives neutrally named Feed additive A and Feed additive B and the opt-out alternative to reflect real-life choices of feeding, where a farmer can typically maintain current feed practices. The opt-out alternative 'None of these. Keep current practices' was included in each decision situation in a similar formulation to previous studies (e.g. Oyinbo and Hansson, 2024). We used simulations of the design data to assess efficiency and needed statistical power for obtaining envisaged priors for a sample size of 240 farmers with 500 draws. The findings suggested that the design recovers truthful estimates for the underlying priors of the simulation with statistical significance for all attributes except for the methane 20 per cent level (simulated P-value .055). We exceeded the sample size of 240, and as the simulated P-value is close to 0.05, our design has sufficient power to detect the effect sizes assumed in the simulation, using the MNL model. Our final sample size also exceeds the minimum sample size of 219 for an MNL

2 Similarly to the recently introduced climate milk-price bonuses from the processors in Sweden, such a bonus would be taxable income. However, we do not anticipate taxation considerations to strongly negatively affect the preference for this attribute since Swedish farmers have various options to optimize their taxation, including tax smoothing across the years (i.e. 'Periodiseringsfond' or Profit Equalization Reserve) and existing survey findings indicate that they have high financial literacy in terms of the use of taxation smoothing practices (Gottlieb and Hansson, 2024).

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model for a 5 per cent significance level and 0.8 power when calculated using an alternative method of de Bekker-Grob *et al.* (2015).

3.3 Split sample treatment

We introduced a policy objectives information treatment randomized among a control and treatment group. The treatment group was shown the text in Box 1. The treatment is meant to examine whether more information, more detailed information, as well as information about public policy objectives such as those of Sweden on top of private policy objectives of the dairy processor companies affects the adoption of feed additives and the preference for the methane-reduction attribute. Therefore, the treatment is not meant to isolate one specific aspect but to offer practice-aligned findings given that several considered aspects of information provision might be offered to farmers. The control group did not receive any information, as in similar information provision experiments (Haaland *et al.*, 2023), including experiments with our target population of Swedish dairy farmers (Oyinbo and Hansson, 2024).

Box 1. Information treatment.

'Various actors in the supply chain, consumers and policy-makers want dairy farms to reduce emissions of greenhouse gases including methane. Dairy companies in Sweden have set science-based targets to reduce greenhouse gas emissions from the farms by about 30% by 2030 to be able to continue selling dairy products to retailers. Moreover, 155 countries have signed up to the **Global Methane Pledge** of reducing methane emissions by 30% by 2030. Sweden has adopted a **Methane Action Plan** that covers agriculture. Agriculture and specifically enteric fermentation of ruminant animals is the major source of methane emissions. The Swedish Environmental Protection Agency (EPA) has subsequently received a task to propose how Sweden can reduce methane emissions, including from agriculture, to meet its commitments under the Global Methane Pledge. In the **report** on this task, the Swedish EPA includes methane-reducing feed additives. Reducing methane emissions is part of the overall Swedish climate targets'.

Note: Bold underlined text included links that respondents could click on, directing them respectively to the website of the GMP.[3] Swedish Government[4] and report of the Swedish EPA.[5]

Respondents spent on average 34 seconds on the treatment text (median 30 seconds), which we view as a sufficient amount of time. Overall, survey

3 https://www.globalmethanepledge.org/#pledges

4 https://www.regeringen.se/contentassets/eba3c73348294c28be87c9b6f0bb4bd5/swedens-methane-action-plan---mapping-of-swedens-methane-emissions-projections-policies-and-measures.pdf

5 https://www.naturvardsverket.se/4a73a2/contentassets/a6f415dca04b468099ecc62bbbe76db3/regeringsuppdrag-minskade-utslapp-av-metan.pdf

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response times are similar across treatment and control groups, reducing concerns about fatigue and inattention. Related to the treatment, we tested for the impact of the information through a manipulation check in the later part of the survey. Namely, we asked the following question 'Both the dairies' research-based GHG targets and the global methane action plan have a goal of reducing emissions by how many %?', with the options 10 per cent, 20 per cent, 30 per cent, 40 per cent and do not know. We randomized this question to appear soon after the choice cards or later for half of the participants. As expected, the Treatment group scores better than control on this question, both when asked early (35 per cent vs. 16 per cent) and later (28 per cent vs. 15 per cent). Section 4 and Table A7 further detail how we considered this variable in robustness analysis.

3.4 Econometric estimation

Farmers' decisions about the use of feed additives are modelled using a random utility maximization framework (McFadden, 1986), allowing us to estimate preferences for feed additive attributes (details in the Section C of the Supplementary Appendix). We estimated MNL and MXL models in utility space with normally distributed parameters for the pooled sample, as well as for treatment and control groups. Based on the base model in utility space, we calculate the WTA using the Delta method. We also estimate the MXL in WTA space, yielding comparable welfare estimates. The treatment effect is captured by an interaction term, and group differences are tested using the Poe test (Poe *et al.*, 1994, 2005). To further explore preference heterogeneity, we estimate a Latent Class Conditional Logit model. For a more detailed explanation of the models and additional robustness checks, see Supplementary Appendix Section C.

3.5 Data

We distributed the survey in May 2024 to all specialized dairy farmers in Sweden that according to the registry of the Statistics Sweden, have an email address or a mobile number. A total of 2,077 farms fulfilled this criterion. We used part of this population for the pilot study, resulting in 57 responses. The survey was distributed by email and/or short messages by a market research company, including two reminders and administering two lottery tickets per completed survey to increase the response rate and prevent survey fatigue of the Swedish farmers. Respondents were invited to take part in the survey about 'feed additives'. Data was anonymous to the researchers. Of the 812 respondents that clicked on the survey link, 391 answered the first question, 360 completed all choice cards and 331 completed the whole survey, suggesting attrition of 15 per cent from the first to last survey question. We removed 14 observations that did not meet the criteria of having dairy cows. The final sample included 320 observations, yielding a response rate of 18.3 per cent, comparable to recent studies of Swedish dairy farmers (e.g. 18 per cent after three reminders in Oyinbo and Hansson, 2024). Median response time to the

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survey was 30 minutes. Table A1 in the Supplementary Appendix shows detailed summary statistics, disaggregated by the treatment and control group, highlighting also differences in covariates. While the number of statistically significant differences is larger than expected with a 5 per cent false positive rate, we have no reason to believe that the process of randomization did not achieve its aim to cut causal backdoors. As argued by Briz *et al.* (2017), an imbalance in covariates is only a concern if these covariates critically interact with the treatment.

Table A2 in the Supplementary Appendix further shows the sample to not be statistically significantly different from the population of Swedish dairy farmers in terms of age and farm size in dairy cow numbers or arable land. While farms are also comparable in the share of organic farms, our sample has somewhat more female and highly educated respondents than in the population, albeit based on estimated and less complete data.

4. Results

Our overall research question and H1-H2 concerned preferences and WTA for feed additive provision attributes as well as their overall uptake potential. Table 2 presents MNL and MXL estimates from the pooled sample of 320 respondents that completed the full survey. While the positive significant estimate for the alternative-specific constant (ASC) suggests that farmers derive higher utility from their current feeding practices than the feed additive, they can be viewed to be nevertheless open to adopting the feed additive given that the opt-out is chosen only 25.6 per cent of the time. In 14.6 per cent of the time, farmers choose an alternative when the offered bonus is zero.

All statistically significant estimates in the MNL model (1) have the expected sign, indicating positive preference for the status quo, higher methane reduction than the 10 per cent baseline, and higher bonus payments. When introducing preference heterogeneity in an MXL model (2), previously non-significant advisory attributes become significant, pointing to preference heterogeneity due to statistically significant standard deviations (third column in Model 2). Models (3–5) present mean WTA calculated from utility space models using the Delta method for the pooled sample and separately for the Treated and Control groups. Note that WTA values in the pooled sample are similar to those calculated from the MNL model (not reported) for the attributes statistically significant in MNL.

The mean WTA shows that to forego the status quo and adopt feed additive without advisory support, the dairy company would have to pay about 8 öre (ca. 0.7 Eurocent) per kg of milk on average. The advisory provision by farm advisors lowers the WTA substantially, but the other two types of advisors have a much lower reduction in WTA. Together with the estimates in (2), this points to a positive preference for all types of advisory providers, aligned with H1. In contrast to our H2, offering more advice increases the WTA and has a negative estimate, indicating that farmers, on average, are not interested

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Table 2. Conditional logit (MNL), MXL and WTA estimates.

	(1) MNL in utility space	(2) MXL in utility space		(3) Pooled sample	(4) Control sample	(5) Treated sample
	Mean est.	Mean est.	SD est.	Mean WTA	Mean WTA	Mean WTA
ASC: status quo	1.08***	1.18***	4.58***	−7.45***	−10.58***	−6.253**
	[7.68]	[2.86]	[9.34]			
Advice: farm advisor	0.25	1.35***	0.93***	−8.53***	−9.199***	−8.199***
	[1.63]	[5.14]	[3.12]			
Advice: veterinarian	−0.06	0.58***	0.61**	−3.64***	−4.856**	−3.000*
	[−0.39]	[2.65]	[2.06]			
Advice: farmer	0.07	0.64***	1.37***	−4.05***	−3.840*	−4.124***
	[0.53]	[3.13]	[5.49]			
Advisory frequency	0.06	−0.22**	0.54***	1.37***	1.150	1.598***
	[1.42]	[−2.51]	[4.62]			
Methane reduction 20 per cent	0.34***	0.87***	0.27	−5.47***	−6.220***	−4.604***
	[3.52]	[5.13]	[0.65]			
Methane reduction 30 per cent	0.43***	1.01***	0.11	−6.38***	−8.530***	−4.870***
	[4.47]	[5.77]	[0.81]			
Methane reduction 40 per cent	0.51***	1.42***	0.88***	−9.00***	−10.75***	−7.670***
	[4.69]	[6.76]	[2.78]			
Bonus, öre per kg milk	0.06***	0.16***	0.14***			
	[12.39]	[9.47]	[8.10]			
ASC: p(WTAc ≤ WTAt)						0.840
Met. 20 per cent: p(WTAc ≥ WTAt)						0.198
Met. 30 per cent: p(WTAc ≥ WTAt)						0.044
Met. 40 per cent: p(WTAc ≥ WTAt)						0.085
Log Likelihood	−1917.16		−1491.79			
Akaike Inf. Crit.	3852.33		3019.59			
Bayesian Inf. Crit.	3912.26		3139.45			
N choices	5760		5760	5760	2790	2970
N respondents	320		320	320	155	165

Notes: * $P < .1$; ** $P < .05$; *** $P < .01$. N respondents 320. N choices 5760. z-values in brackets for (1) and (2). Standard errors clustered at respondent level. The MXL model uses 1,000 Halton draws for the full sample and 1,500 Halton draws for the Treated and Control samples. WTA estimated using the Delta method. Comparison of the WTA for the Treated and Control sample ares done using Poe test.

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in receiving advice more frequently and would like to avoid it. Even though we mentioned that advice relating to the frequency attribute is offered free of charge and may allow for answering emerging questions with the latest information, there is, nevertheless, the opportunity cost of labour to engage with the advisor, and it may be higher than the perceived value of advice. A positive preference towards higher effectiveness of the feed additives also results in more emission reduction, substantially lowering the WTA.

As a robustness test, we re-estimated the models with all respondents who completed the choice tasks, but there were no larger differences in estimates (Table A3 in the Supplementary Appendix). We also re-estimated the models, excluding eight farmers who had tested the feed additive in a pilot, which yielded comparable results (Table A4 in the Supplementary Appendix).

Regarding the role of information treatment on WTA and H3-H4, we investigate the treatment effects of the information script by applying the Poe test to compare WTA estimates between the treated and control samples (Poe *et al.*, 1994, 2005). Table 2 shows that the differences in WTA are marginally significant at 5 per cent and 10 per cent for the highest methane reduction potential attribute levels, 30 per cent and 40 per cent, respectively. The effect on the 30 per cent level is intuitive, considering that treatment information emphasized this emission reduction goal. We thus find some support for the H3. We do not find support for H4 regarding the negative treatment effect on the opt-out. Detailed estimates from the underlying utility space preference parameters and WTA confidence intervals for the treated and control samples are presented in Table A5 in the Supplementary Appendix. Estimates presented in Table A6 interact the treatment with the attributes but do not show statistically significant interaction terms. We also tested the model with three-way interaction terms, involving the attributes, Treatment and answering correctly to the manipulation check (Table A7 in the Supplementary Appendix). This also did not yield any relevant statistically significant interaction terms.

Finally, exploring what behavioural factors drive farmer decisions in this context, we considered several options to model observed and unobserved heterogeneity. The MXL models suggest unobserved heterogeneity, but it is not obvious how to best model this unobserved heterogeneity and how to incorporate covariates for modelling observed heterogeneity. An alternative to the continuous distribution of the MXL model is LCCL, which is suitable for exploring preference heterogeneity among latent classes of respondents. In addition, it allows for a more parsimonious model by including covariates in the membership function, avoiding a large number of interactions of covariates and attributes. We explored different numbers of classes, from two to five, where the two-class model provided the best fit based on the Bayesian Information Criterion (see Table A8 for an overview of goodness of fit measures for the different estimated models). Table 3 shows estimation results.

The two-class model separates the probability of respondents belonging to one of two groups: (1) Sceptics—a conservative group with a strong preference for keeping the status quo that does not like to adopt a feed additive; and (2) Potential adopters—a group of farmers that is likely to adopt the feed ad-

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Table 3. Coefficient estimates of the two-class LCCL model.

	Class 1 Sceptics		Class 2 Potential adopters	
	Est.	Std. error	Est.	Std. error
Average class probability	17.7 per cent		82.3 per cent	
ASC: status quo	4.93***	(0.885)	0.54***	(0.195)
Advice: farm advisor	0.51	(0.833)	0.94***	(0.203)
Advice: veterinarian	−0.42	(0.897)	0.41**	(0.177)
Advice: farmer	−0.12	(0.828)	0.49***	(0.170)
Advisory frequency	0.13	(0.250)	−0.20***	(0.060)
Methane reduction 20 per cent	0.86	(0.556)	0.56***	(0.118)
Methane reduction 30 per cent	0.80	(0.550)	0.60***	(0.121)
Methane reduction 40 per cent	0.25	(0.590)	1.09***	(0.128)
Bonus, öre per kg milk	0.09***	(0.023)	0.10***	(0.007)
Class membership				
Age	0.06**	(0.027)	–	
Female	1.24**	(0.606)	–	
Post-primary education	−0.58	(0.536)	–	
Conventional	1.57**	(0.798)	–	
General trust	0.01	(0.561)	–	
Confidence in stakeholders	−0.34	(0.407)	–	
Innovativeness factor score	−0.90**	(0.391)	–	
Risk-taking	0.42***	(0.158)	–	
Prominence of FAW views	1.04***	(0.366)	–	
Raises bulls or steers	−0.71	(0.560)	–	
Has feeding stations	0.37	(0.532)	–	
Has a feed mixer	0.36	(0.589)	–	
Several production/feeding systems	−0.83	(0.539)	–	
Biogas production	2.29***	(0.858)	–	
% others ready to test	−0.06***	(0.017)	–	
Tested feed additive	−0.73	(1.312)	–	
Delivers to Norrmejerier	−1.12	(0.887)	–	
Delivers to Arla	−1.70**	(0.726)	–	
Sustainable feeding practices	0.19	(0.191)	–	
Familiar factor score	0.59*	(0.321)	–	
Extra trial information important	−2.31***	(0.641)	–	
Constant	−6.07***	(2.351)	–	

Notes: *** $P < .01$, ** $P < .05$, * $P < .1$. Standard errors in parentheses. The analysis is based on 4,356 choices from 242 respondents. The lower number of observations compared to the choice models without covariates is due to missing data for some of the covariates.

ditive, especially when advice is provided and when the effectiveness is high. Notably, the average probability of belonging to the conservative group is substantially smaller (17.7 per cent). In other words, there is a higher than 80 per cent probability that a farmer in our sample belongs to the group open to feed

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additives. The groups do not substantially differ in utility from the bonus payment, but the utility from other attributes is much lower overall. To overcome the status quo barrier, much higher payments must be made to the conservative group.

Class membership estimates for class 1, the 'Sceptics', suggest that these farmers are more likely to be older, have conventional and biogas production and prioritize FAW more strongly against environmental and economic considerations. Similar to the survey findings from Finland, women are more likely to oppose feed additives (Sidhoum *et al.*, 2025). In other words, scepticism of this class may emerge from the relative novelty of feed additives and concerns regarding the impact of additives on manure for biogas production or FAW. We asked farmers about their expectations of other farmers' willingness to adopt feed additives. Those holding more pessimistic beliefs (variable '% others ready to test') are also more likely to fall into the conservative group. This highlights the role of peer imitation and network effects, consistent with modelling of 3-NOP mitigation potential in voluntary carbon markets (Tarruella *et al.*, 2025). In line with the literature (Barreiro-Hurle *et al.*, 2023; Schaub *et al.*, 2023), behavioural factors such as norms and risk preferences drive the probability of class membership. Interestingly, willingness to take risks was associated with *not* adopting the new technology. One possible explanation could be that the risk-taking measure we used is general, and future studies may wish to validate this finding with a more domain-specific risk measure.

Class 2, the 'Potential adopters', representing 82.3 per cent of the sample on average, includes farmers with statistically significantly positive preferences towards attributes of the offered feed additives. This class has the strongest positive prefere*nce for* advice from farm advisors, similarly to the MXL results earlier. Moreover, this class has a clear preference for the methane-reduction attribute, with a stronger p*ositive preference for* greater reduction levels. We can further note that membership in class 2 is increased by being more innovative, being interested in information from long-term trials on feed additives, and delivering milk to Arla, Sweden's largest dairy processor, which has run pilots on feed additives.

Finally, based on the estimates for the treated and control samples in the MXL models, we estimate the market shares of adopting the feed additive across various bonus levels for an additive with a 10 per cent vs. 30 per cent methane reduction efficiency and accompanied by advice from a farm advisor once a year vs. no advisory. Figure 2 shows very high expected adoption even without any bonus on top of covering the cost of the feed additive, especially for the commercially claimed 30 per cent reduction: at zero bonus, the supply curves already start at approximately 40 per cent for implementation without advisory and approximately 70 per cent for adoption accompanied by a farm advisor. Even the supply curves for an additive with a 10 per cent reduction begin at non-trivial adoption rates without a bonus. Both methane efficiency and advisory support shift each curve upward at any bonus level, increasing the take-up. Comparing panels A and B, the treat-

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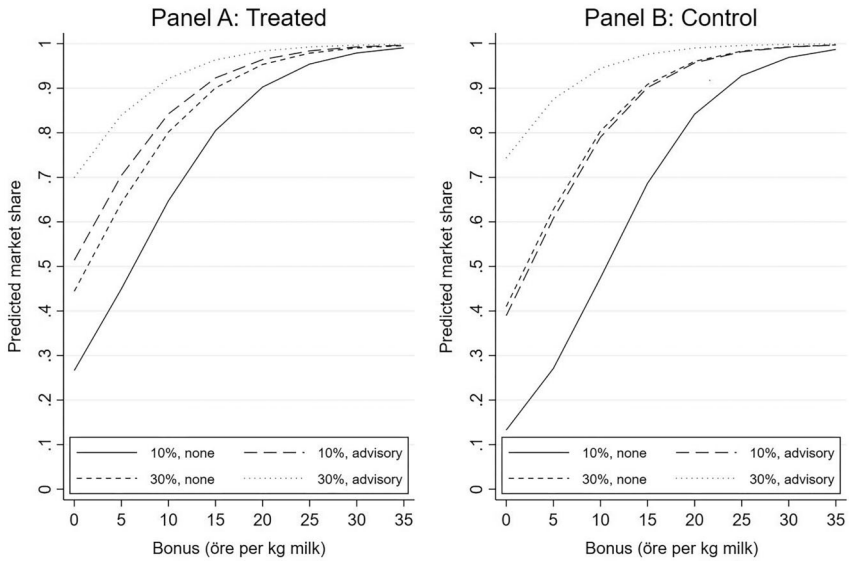

Fig. 2. Supply curves for feed additive adoption by information treatment.
Notes: Supply curves are based on the MXL estimates for the treated and control samples, respectively, in panels A and B. The supply curves in each panel depict adoption packages based on either 10 per cent or 30 per cent methane reduction efficiency of the feed additive and whether farm advisory once a year is offered, compared to no advisory provision.

ment effect of policy target information is modest: the treated group reaches a given adoption level at slightly lower bonuses than the control, but the gap is small relative to the much larger differences due to methane efficiency and offered advice. As the bonus increases, all curves quickly converge toward near-universal adoption by roughly 25–30 öre/kg (ca. 2.5–3 Eurocent/kg).

To put these economic results in a broader perspective, consider Table 4, which shows the emission abatement costs per ton of carbon equivalent under the 30 per cent methane emission reduction efficiency, representing the commercially claimed methane reduction efficiency of *Bovaer*. Given the heterogeneity in emission intensity (E) of different milk production systems, we also consider a 1 and 1.4 kg CO_2-equivalent emission per kg of milk scenario, which is akin to the intensive production in Sweden and other Western European countries, as well as a 3.7 kg scenario for more extensive production globally.

The table shows that for a low cost of implementing feed additives per kg of milk, the marginal abatement cost is relatively low under this emission scenario (assuming no advisory support and costs comprised of only the bonus and the cost of the feed additive). The current cost of the feed additive *Bovaer* is estimated to be about one Eurocent per kg of milk. Allowing for an additional 10 öre/kg (0.8 Eurocent/kg) for bonus payments, even for the most

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Table 4. Treatment effect on marginal abatement costs per ton of carbon equivalent for different adoption rates and emission reduction scenarios.

Adoption target	Group	Required bonus (öre/kg milk)	Required bonus (Eurocent/kg)	Total cost per kg for MAC (Eurocent) = 1.0 + bonus	MAC at $E = 1.0$ (Euro/t)	MAC at $E = 1.4$ (Euro/t)	MAC at $E = 3.7$ (Euro/t)
40 per cent	Treated	0.0	0.0	1.0	74.1	52.9	13.9
40 per cent	Control	0.0	0.0	1.0	74.1	52.9	13.9
60 per cent	Treated	3.9	0.3	1.3	98.1	70.1	18.4
60 per cent	Control	4.4	0.4	1.4	101.2	72.3	18.9
80 per cent	Treated	10.0	0.8	1.8	135.8	97.0	25.4
80 per cent	Control	9.9	0.8	1.8	135.2	96.6	25.3
99 per cent	Treated	29.8	2.5	3.5	258.0	184.3	48.3
99 per cent	Control	28.0	2.3	3.3	246.9	176.4	46.2

Notes: The exchange rate assumed is 12 SEK/Euro, as was the case during the data collection. The cost of the feed additive covered for farmers is assumed to be 1 Eurocent. The share of methane enteric fermentation emissions in total CO2e emissions is assumed to be 0.45 for 1–1.4 emission intensity scenarios (Mazzetto *et al.*, 2022), and 0.65 for the 3.7 emission intensity scenario.

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efficient farms, the estimated abatement cost is only about 135 Euro per ton of carbon for a substantial adoption rate of 80 per cent, which only slightly exceeds the current carbon tax in Sweden for fossil fuels. At near-complete adoption, the MAC increases to approximately 250 Euro. While adopting the additives on extensive farms with high emissions per kg of milk could lower these costs, attention needs to be paid to the potential role of the farm structure on WTA (Luke and Tonsor, 2024). The information treatment appears to have a modest positive effect at lower bonus levels and negative at higher ones. The low estimated cost of the feed additive required for broad adoption is attractive when compared to measures with high up-front cost, complex infrastructure and longer timelines such as carbon farming (Block *et al.*, 2024) or more complex agri-environmental policies relying on more fundamental changes to farming practices for which farmers often demand high payments (Schulze *et al.*, 2024).

5. Concluding remarks

Our study examined the role of two distinct types of information strategies—advisory provision accompanying implementation and ex-ante policy information—in shaping Swedish dairy farmers' preferences and WTA methane-reducing feed additives. First, the results show that advisory support matters: who provides the advice and how often it is given both influence uptake potential. In line with earlier evidence that context-relevant, complementary, and start-up advisory support can facilitate adoption decisions (e.g. Zemo and Termansen, 2018; Späti *et al.*, 2022; Kirchner and Musshoff, 2024), our findings highlight that advisory provision helps to reduce WTA for the uptake. Similar to the findings in the US context (Tynan *et al.*, 2023), farm advisors, who in Sweden also heavily advise on livestock nutrition, reduce WTA more than veterinarians and other farmers. This warrants continuing with the approach tested in pilot trials on farms in Sweden, where a farm advisor was involved. However, the unexpected results about the negative preference for advisory frequency suggest that more visits are counterproductive to farmers, as farmers perhaps do not expect the value of more updated information to offset the opportunity cost of their time. Latent class analysis shows heterogeneity of the preferences for advisory provision, with a majority of farmers for whom advisory provision is relevant being open to information from long-term trials and, in general, more innovative. While we find the smaller 'sceptical' latent class of farmers to indeed hold stronger pro-FAW views, this does not appear to be relevant for their preference for veterinarians as providers of advisory, perhaps as the implications of the additives may be assumed to be linked to digestive (and thus farm advisor-related) rather than other health considerations.

For the policy information treatment, the results of the Poe test suggest that the information provision had a 5 per cent statistically significant effect on WTA estimates for the 30 per cent methane-reduction level—in focus of the

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treatment—and a marginally significant effect at the 40 per cent level. However, the absence of statistically significant treatment–attribute interactions tempers this conclusion. This finding aligns with the mixed evidence in the literature on how information affects the adoption of new farming practices (e.g. Zachmann *et al.*, 2023; Oyinbo and Hansson, 2024) and on behavioural interventions more generally (Maier *et al.*, 2022; Szaszi *et al.*, 2022). While the positive effect of information does not hold in all models, we also did not find any indication of negative effects, such as psychological reactance (Bruns and Perino, 2023), which can be a concern in behavioural policy, particularly in agriculture (Colen *et al.*, 2024; Finger *et al.*, 2024). That being said, treatment effects in this context may be small and sensitive to how messages are framed, and our study design does not allow detection of very small effects with high confidence, echoing the insights of Barreiro-Hurle and Rommel (2025). Given that farmers in previous work have expressed uncertainty about the effectiveness of feed additives (Sidhoum *et al.*, 2025), future research should use larger, more diverse samples to test alternative ways of framing such information, perhaps alongside information on *long-term* effectiveness and effects, including on FAW.

Taken together, in practical terms, the results suggest that information provision to accompany feed additives adoption may warrant concise start-up advisory information by farm advisors, ideally incorporating lessons from the rollout of feed additives in Denmark to emphasize the new social norms and especially the findings from the application of the additives by farmers that opt to use them above the minimum required duration. Overall, our study suggests that the adoption of methane-reducing feed additives in Sweden could be substantial, even at low bonus levels, if cost coverage is assured, as in Denmark. While we find the content of the ex-ante information treatment about methane reduction target levels to be relevant for the most related attribute of methane efficiency, the effect of information aligned to a relevant attribute is still rather modest and may be viewed as merely complementary.

Our study has at least three limitations. First, we focus only on Sweden, a country with rather extensive farming practices by Western European standards and rather intensive farming practices by global standards. Sweden also has unique social and cultural values (Colen *et al.*, 2024). Hence, it would be important to replicate our results in other countries and closely follow the actual adoption of feed additives. Second, our sample is relatively small. Although our data covers approximately one-fifth of Sweden's dairy farms and is representative for several important farm characteristics, several of the models we estimated have many parameters, and the limited data does not allow us to estimate more demanding choice models such as hybrid choice models. Hybrid choice models would also allow for a greater consideration of behavioural aspects such as fairness or identity (Zemo and Termansen, 2022). Finally, there is still uncertainty regarding the investigated feed additives' long-term effectiveness and possible side effects (Arndt *et al.*, 2022; Gastelen *et al.*, 2024). Ultimately, we only begin to understand the potential of this new technology. As of today, our results allow a positive conclusion. Feed additives will

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probably be widely adopted, not least because they seemingly offer a simple solution to a complex problem, taking away some pressure from farmers and dairy processors to deliver on climate targets.

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Supplementary data

Supplementary data are available at *ERAE* online.

Conflicts of interest. None declared.

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Data availability

Materials, data and code are available with the submission.

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